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PROBABILITY & STATISTICS

MODULE - 1

Time Series Analysis

$$y = f(t) \begin{matrix} \nearrow \text{time} \\ \downarrow \text{phenomena (variable)} \end{matrix}$$

A time series is an arrangement of statistical data in a chronological data i.e., in accordance with its time of occurrence.

Eg:- The series relating to prices, production and consumption of various commodities, agricultural and industrial production, investment, sales and profits of business concerns, bank deposits and bank clearings, etc are all time series spread over a long period of time.

Mathematically, time series is defined by the functional relationship $y = f(t)$ where y is the value of the phenomena (or variable) under consideration at a time ' t '.

- Eg:-
- i) Population Y of a country or a place in different years ' t '.
 - ii) The number of births and deaths in different months ' t ' of the year.

Components of a time series. If the values of a phenomena are observed at different periods of time, the values so obtained will show changes, these fluctuations are due to the

fact that the value of the phenomena or variable is affected not by the single factor but due to the cumulative ~~affects~~ effects of a multiplicity of factors pulling it up & down.

Eg. The sales (y) of a product are influenced by :-

- 1) Advertisement expenditure
- 2) The price of the product
- 3) The income of the people
- 4) Other competitive products in the market & so on

The time series data contains 4 components :-

- 1) Trend
- 2) Cyclical variation
- 3) Seasonal variations
- 4) Irregular variations

PROBLEMS

Measurement of Trend

→ The following are the 4 methods which are generally used for the study & measurement of trend components in a time series.

- 1) Graphic method
- 2) Method of Semi-average
- 3) Method of curve-fitting by the principle of least squares
- 4) Method of moving averages

7) Fit a linear trend to the following data by the least squares method. Verify that $\sum(y - y_e) = 0$ where $y_e \rightarrow$ corresponding trend value of y .

Year (time)	1990	1992	1994	1996	1998
Production in (1000 units)	18	21	23	27	16

Also estimate the production for the year 1999.

Sol	Year	Production (y)	$x = t - 1994$	x^2	xy	$y_e = a + bx$ $y - y_e$
	1990	18	-4	16	-72	20.6
	1992	21	-2	4	-42	20.8
	1994	23	0	0	0	21
	1996	27	2	4	54	21.2
	1998	16	4	16	64	21.4
	$\Sigma = 105$		$x = t - 1994$	$\Sigma = 0$	$\Sigma = 40$	$\Sigma = 4$

when odd no. of data is given take middle value

$$y_e = a + bx$$

$$\Sigma y = 5a + b \Sigma x \rightarrow (1)$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \rightarrow (2)$$

On solving (1) & (2)

$$105 = 5a + b(0)$$

$$a = \frac{105}{5}$$

$$a = 21$$

$$4 = 21(0) + b(40)$$

$$40b = 4$$

$$b = 4/40$$

$$b = 0.1$$

$$y_e = 21 + 0.1(x)$$

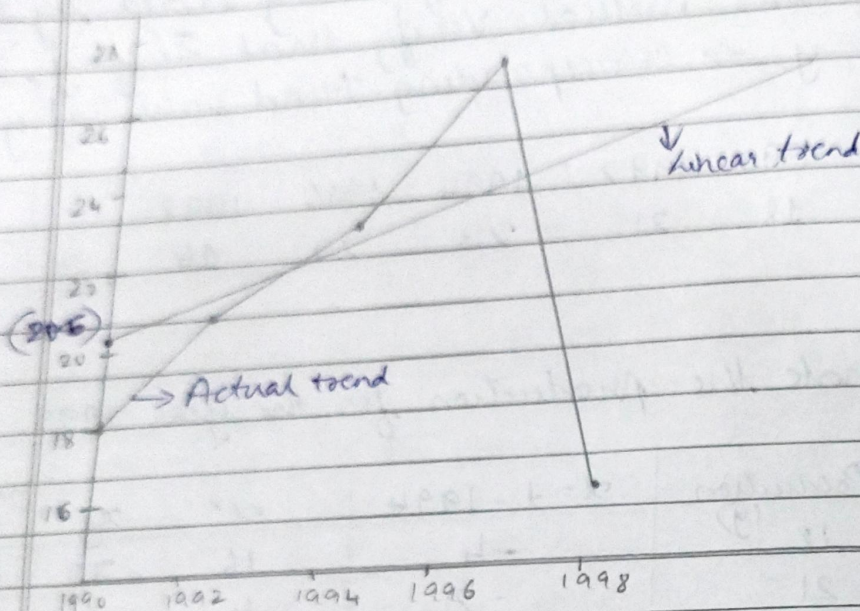
↑ trend is increasing
↑ slope

When $t = 1999$

$$x = 1999 - 1994$$

$$x = 5$$

$$\therefore y_e = 21 + 0.1(5) = 21.5$$



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2) Below are given the figures of production (in thousand tons) of a sugar factory.

Year (t) :	1999	2000	2001	2002	2003	2004	2005
Production (y) :	77	88	94	85	91	98	90

- Fit a straight line by the method of least squares and show the trend values.
- What is the monthly increase in production?
- Eliminate the trend by using both additive and multiplicative model.

Sol

$$x = t - 2002$$

odd no of data, take middle value.

$$y = a + bx$$

∴ The Normal equation is given by

$$\sum y = 7a + b \sum x \quad \& \quad \sum xy = a \sum x + b \sum x^2$$

$\hookrightarrow \textcircled{1}$
 $\hookrightarrow \textcircled{2}$

Year (t)	Production (y)	$x = t - 2002$	xy	x^2	$y_e = 89 + 2x$ (predicted) 8.3 trend value
1999	77 (actual)	-3	-231	9	83
2000	88	-2	-176	4	85
2001	94	-1	-94	1	87
2002	85	0	0	0	89
2003	91	1	91	1	91
2004	98	2	196	4	93
2005	90	3	270	9	95
	<u>623</u>	<u>0</u>	<u>56</u>	<u>28</u>	

From ① & ②

$$623 = 7a + b(0)$$

$$56 = a(0) + b(28)$$

$$a = \frac{623}{7}$$

$$\boxed{a = 89}$$

$$\frac{56}{28} = b$$

$$\boxed{b = 2}$$

if b is +ve
trend is
increasing

$$\boxed{y = 89 + 2x}$$

(thousand tons)

$$\text{Yearly increment} = 2 \times 1000$$

$$= 2000 \text{ units}$$

$$\text{Monthly increment} = \frac{2000}{12} = \underline{\underline{166.66 \text{ units}}}$$

Production (y)	y_e	Additive model $[y - y_e]$	Multiplicative model y/y_e
77	83	-6	0.927
88	85	3	1.035
94	87	7	1.0804
85	89	-4	0.955
91	91	0	1
98	93	5	1.053
90	95	-5	0.947

3) The sales of a company in million rupees for the year 1994-2001 are given below.

Year	1994	1995	1996	1997	1998	1999	2000	2001
Sales	550	560	555	585	540	525	545	585

- Find the linear trend equation
- Estimate the sales for the year 1993.
- Find the slope of the straight line trend.
- Do the figures show a rising trend or a falling trend?

NOTE : When n is even

$$x = t - \frac{(\text{Arithmetic mean})}{\frac{1}{2}(\text{interval } (h))}$$

When n is odd

$$x = t - \frac{(\text{middle value})}{\text{Interval } (h)}$$

Sol.

$$x = t - \frac{(1997+1998)}{2}$$

$$x = t - 1997.5$$

~~$x = t - 1997.5$~~

$$x = 2(t - 1997.5)$$

year(t)	sales(y)	$x = 2(t - 1997.5)$	xy	x^2	$y_e = 555.625 + 0.2083x$
1994	550	-7	-3850	49	554.1669
1995	560	-5	-2800	25	554.5835
1996	555	-3	-1665	9	555.0001
1997	585	-1	-585	1	555.4167
1998	540	1	540	1	555.8333
1999	525	3	1575	9	556.2499
2000	545	5	2725	25	556.6665
2001	585	7	4095	49	557.0831
	<u>4445</u>	<u>0</u>	<u>35</u>	<u>168</u>	

$$y = a + bx$$

$$\sum y = 8a + b\sum x \rightarrow \textcircled{1}$$

$$\sum xy = a\sum x + b\sum x^2 \rightarrow \textcircled{2}$$

From ① & ②

$$4445 = 8a + 0$$

$$a = \frac{4445}{8} = \underline{\underline{555.625}}$$

$$35 = b(168)$$

$$\frac{35}{168} = b$$

$$b = \underline{\underline{0.2083}}$$

i) $\boxed{y = 555.625 + 0.2083x}$ slope $\rightarrow$ Trend value

ii) $x = 2(1 - 1997.5)$
 $x = 2(1993 - 1997.5)$
 $\boxed{x = -9}$

$$\boxed{y_e = 553.75}$$

iii) 0.208

iv) Rising trend as slope is positive

4) Calculate the quarterly trend values by the method of least squares for the following quarterly data for the last 5 years.

Year	I st Quarter	II Quarter	III Quarter	IV Quarter
1994	60	80	72	68
1995	68	104	100	88
1996	80	116	108	96
1997	108	152	136	124
1998	160	184	172	164

$$(60+80+72+68)/4$$

Sol

Year(t)	Average of Quarter Value(y)
1994	70
1995	96
1996	100
1997	130
1998	170
	<u>560</u>

$x=t-1996$	x^2	xy	$y_e = 112 + 24x$
-2	4	-140	64
-1	1	-90	88
0	0	0	112
1	1	130	136
2	4	340	160
<u>0</u>	<u>10</u>	<u>240</u>	

$$y = a + bx$$

$$\sum y = 5a + \sum bx$$

$$\sum xy = a \sum x + b \sum x^2$$

$$560 = 5a$$

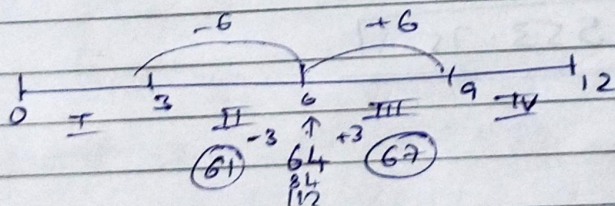
$$a = \frac{560}{5} = \underline{\underline{112}}$$

$$240 = 10b$$

$$\underline{\underline{b = 24}}$$

$$\underline{\underline{y = 112 + 24x}}$$

$$\text{Quarterly increment} = \frac{24}{4} = 6$$



Year	I Quarter	II Quarter	III Quarter	IV Quarter
1994	55	61	67	73
1995	79	85	91	97
1996	106 103	109	115	121
1997	127	133	139	145
1998	151	157	163	169

MOVING AVERAGE METHOD

A quantitative method of forecasting a time series by averaging each successive group of data values

1) Calculate 3 yearly, 5 yearly moving average for the following data

year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
y	242	240	252	249	253	255	251	257	260	265	262

<u>sol</u>	year	y	3 year moving total	3 year moving avg	5 year moving total	5 year moving avg
	1990	242	—	—	—	—
	1991	240	744	248	1246	249.2
	1992	252				
	1993	249				
	1994	253	757	252.33	1260	252
	1995	255	759	253	1265	
	1996	251	763	254.33	1276	
	1997	257	768	256	1288	
	1998	260	782	260.66	1295	
	1999	265	787	262.33	—	—
	2000	262	—	—	—	—

2) Assume a 4 year cycle and the trend by method of moving average from following data relating to production of India.

year	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
x	464	515	518	467	402	540	557	571	586	612

Weighted moving Average - A quantitative method of forecasting a time series by computing a weighted average of data values

→ only for 4 year cycle take
avg $\neq 2$

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year	t	4 year moving total	4 year moving avg	Centric
1987	464			
1988	515	1964	491	
1989	518			483.25
1990	467	1902	475.5	
1991	402	1927	481.75	478.63
1992	540	1966	491.5	486.63
1993	557	2070	517.5	504.5
1994	571	2254	563.5	540.5
1995	586	2326	581.5	572.5
1996	612			

3) A vacuum cleaners sales for 12 months is given below

Month	1	2	3	4	5	6	7	8	9	10	11	12
Actual sale	10	12	13	16	19	23	26	30	28	18	16	14

Sl	Month	Sales (y)	3 months moving total	3 months moving avg
	1	10	—	—
	2	12	35	11.67
	3	13	41	13.67
	4	16	48	16
	5	19	58	19.33
	6	23	68	22.67
	7	26	79	26.33
	8	30	84	28
	9	28	76	25.33
	10	18	62	20.67
	11	16	48	16
	12	14	—	—

(this is diff problem)
Written later

Seasonal variations

Variations occur in a regular or periodic manner due to natural forces or man-made convention is known as seasonal variation.

Measurement of seasonal variation.

- 1) Method of simple average (x)
- 2) Ratio to Trend method
- 3) Ratio to Moving average
- 4) Link Relative method. (SLT)

Problems

(10-12 marks) Quarterly sales data (Rs in million) in a super bazas are presented in the following table for a 4 year period.

Year	I	II	III	IV
2000	60	80	72	68
2001	68	104	100	88
2002	80	116	108	96
2003	108	152	136	124
2004	160	184	172	164

Calculate the seasonal index for each of the 4 quarters using Ratio to Trend method.

<u>Sol</u>	<u>Year (t)</u>	<u>Total sales (y)</u>	<u>x = t - 2002</u>	<u>x²</u>	<u>xy</u>	<u>y_e = 112 + 24x</u> (Trend values)
	2000	280/4 = 70	-2	4	-140	64
	2001	360/4 = 90	-1	1	-90	88
	2002	400/4 = 100	0	0	0	112
	2003	520/4 = 130	1	1	130	136
	2004	680/4 = 170	2	4	340	160
		<u>Σ = 560</u>	<u>0</u>	<u>10</u>	<u>Σ xy = 240</u>	

$$y = a + bx$$

$$\Sigma y = 5a + b \Sigma x \rightarrow (1)$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 \rightarrow (2)$$

Substituting in ① & ②.

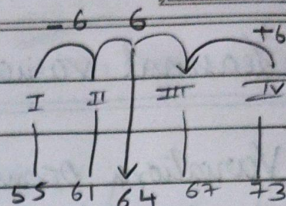
$$560 = 5a$$

$$a = 112$$

$$240 = 10b$$

$$b = 24$$

$$y = 112 + 24x$$



Yearly increment $\Rightarrow b = 24$

Quarterly increment $\Rightarrow 24/4 = 6$

Trend values

	I	II	III	IV
2000	55	61	67	73
2001	79	85	91	97
2002	103	109	115	121
2003	127	133	139	145
2004	151	157	163	169

For demand to 2 Ratio - to - Trend

$$\text{ratio to trend} = \frac{\text{Old value}}{\text{New value}} \times 100$$

Year	I	II	III	IV
2000	109.09	131.15	107.46	93.15
2001	86.08	122.35	109.89	90.72
2002	77.67	106.42	93.91	79.34
2003	85.04	114.29	136 97.84	85.52
2004	105.96	117.20	105.52	97.04
Total	463.84	591.41	552.78 514.62	445.77
Average	92.77	118.28	110.56 102.92	89.15
Seasonal variate Index	<u>92.04</u>	<u>117.36</u>	<u>102.12</u>	<u>88.46</u>

$$\frac{92.77 \times 100}{100.78}$$

$$\text{Seasonal index} = \frac{\text{Quarterly Average}}{\text{General Average}} \times 100$$

$$\text{General Average} = \frac{92.77 + 118.28 + 102.92 + 89.15}{4}$$

$$= 100.78$$

2) Following table gives the sales figures in hundred units for a product.

Year	Quarter			
	I	II	III	IV
1982	36	34	38	32
1983	38	48	52	42
1983				
1984	42	56	50	52
1985	56	74	68	62
1986	82	90	88	80

Use least square method to form equation and compute the demand for the product for the quarter wise using ratio to trend method.

3) Using 'Ratio to Trend' method, determine the quarterly seasonal indices for the following data

Production of coal (in million of tons)				
Year	I Qtr	II Qtr	III Qtr	IV Qtr
1	68	60	61	63
2	70	58	56	60
3	68	63	68	67
4	65	56	56	62
5	60	55	55	58

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Ratio to Moving Average

- i) Calculate seasonal indices by 'ratio to moving average' from the following data of sales (y) in lakh of rupees.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2001	68	62	61	63
2002	65	58	66	61
2003	68	63	63	67

Year	Sales (y)	4 quarterly total	4 quarterly average	4 quarterly average cental	Ratio to M.A value
2001	I (68)	—	—	—	—
	II (62)	254	63.50	63.13	96.63
	III 61	251	125.5 62.75	62.25	101.20
	IV 63	247	61.75	62.38	104.20
2002	I 65	252	63.00	62.75	92.43
	II 58	250 250	62.50	62.88	104.96
	III 66	253	63.25	63.88	95.49
	IV 61	258	64.50	64.13	106.03
2003	I 68	255	63.75	64.50	97.67
	II 63	261	65.25	—	—
	III (63)	—	—	—	—
	IV (67)	—	—	—	—

Computation of Seasonal Index

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2001	—	—	96.64	101.20
2002	104.20	92.43	104.96	95.49
2003	106.03	97.67	—	—
Total	210.23	190.10	201.60	196.69

(Q.2)

Average	105.12	95.05	100.80	98.35
Seasonal index	<u>105.30</u>	<u>95.21</u>	<u>100.97</u>	<u>98.52</u>
	(best season for sales)			

$$\text{General average} = \frac{105.12 + 95.05 + 100.80 + 98.35}{4}$$

$$= \underline{\underline{99.83}}$$

$$\text{Seasonal index} = \frac{\text{Quarterly avg}}{\text{General avg}} \times 100$$

Weighted moving average - A quantitative method of forecasting a time series by computing a weighted avg of past data values.

- 1) Vacuum cleaner sales 'S' for 12 months is given below. The owner of the supermarket decides to forecast sales by weighting the past three months as follows.

Weights applied				Month								
3				last month								
2				2 months ago								
1				3 months ago								
Month	1	2	3	4	5	6	7	8	9	10	11	12
Actual sales	10	12	13	16	19	23	26	30	28	18	16	14

Sol
 since its 3 months ago take from 4th month.
 $3 + 2 + 1 = 6$ so take $1/6$.

Month	Sales	3 month weighted moving avg
1	10	
2	12	
3	13	
4	16	$\rightarrow \frac{1}{6} [3 \times 13 + 2 \times 12 + 1 \times 10] = 12.166$
5	19	$\rightarrow \frac{1}{6} [3 \times 16 + 2 \times 13 + 1 \times 12] = 14.33$
6	23	$\rightarrow \frac{1}{6} [3 \times 19 + 2 \times 16 + 1 \times 13] = 17$
7	26	$\rightarrow \frac{1}{6} [3 \times 23 + 2 \times 19 + 1 \times 16] = 20.5$
8	30	$\rightarrow \frac{1}{6} [3 \times 26 + 2 \times 23 + 1 \times 19] = 23.83$
9	28	$\rightarrow \frac{1}{6} [3 \times 30 + 2 \times 26 + 1 \times 23] = 27.5$
10	18	$\rightarrow \frac{1}{6} [3 \times 28 + 2 \times 30 + 1 \times 26] = 28.33$
11	16	$\rightarrow \frac{1}{6} [3 \times 18 + 2 \times 28 + 1 \times 30] = 23.33$
12	14	$\rightarrow \frac{1}{6} [3 \times 16 + 2 \times 18 + 1 \times 28] = 18.66$

2) Calculate the seasonal index for the following data by using Ratio to moving average method.

Year	Q1	Q2	Q3	Q4
1976	30	81	62	119
1977	33	104	86	170
1978	42	153	99	221